

Nitish Kumar Sharma

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Professional Summary

Final-year B.Sc. Information Technology student with strong foundations in full-stack web development, networking infrastructure, cloud DevOps, and AI automation. Demonstrated ability to architect, deploy, and deliver end-to-end production-grade systems — including a real-time digital learning platform, CI/CD pipelines on AWS, and automated data-analysis agents. Seeking a software engineering or cloud engineering role where I can apply technical depth and a proven record of building scalable systems.

Technical Skills

Languages:	C, C++ (Core), Java, JavaScript (ES6+)
Frontend:	React.js, HTML5, CSS3, Responsive UI Design
Backend & APIs:	Node.js, Express.js, REST API, Socket.IO, WebSockets
Databases:	MongoDB, Microsoft SQL Server, Mongoose ODM
Cloud & DevOps:	AWS (EC2, S3), GitHub Actions (CI/CD), Docker (fundamentals), Linux/Ubuntu
Networking:	DHCP, VLAN, Inter-VLAN Routing, OSPF, NAT, SSH, Port Security, Subnetting — Cisco Packet Tracer
AI & Automation:	n8n Workflow Automation, AI Agent Design, LLM Integration
Tools & Platforms:	Git/GitHub, VMware, VS Code, Power BI, MS Office, Cloudinary
Concepts:	System Analysis & Design, Client-Server Architecture, Data Warehousing, CCNA fundamentals

Projects

Digital Learning Platform System | Demo

React, Node.js, Express.js, MongoDB, Socket.IO, Cloudinary

2024 – 2025

- Architected and deployed a full-stack, real-time digital learning platform from the ground up, supporting concurrent sessions for students, teachers, and administrators across three distinct role-based dashboards.
- Engineered a live classroom module using **Socket.IO** and WebSockets, enabling bidirectional real-time chat, doubt resolution, and interactive lectures with sub-100ms latency.
- Built a complete online examination engine featuring timed assessments, automated grading, instant result computation, and tamper-resistant submission workflows.
- Integrated an **AI-powered chatbot** for on-demand academic assistance, reducing teacher response load and improving student self-service resolution rate.
- Implemented a course-selling and enrollment system with **Cloudinary**-managed media storage, structured course content delivery, and role-based content access controls.
- Designed a normalized **MongoDB** schema across 8+ collections, optimizing query performance through indexing and aggregation pipelines.

Cloud DevOps Pipeline | AWS EC2, GitHub Actions, CI/CD, Docker

2025

- Designed and implemented a fully automated **CI/CD pipeline** using GitHub Actions for a Node.js To-Do application, reducing manual deployment effort to zero.
- Provisioned and configured an **AWS EC2** instance as the production host, establishing secure SSH access, environment variable management, and self-hosted runner integration.

- Achieved automated build validation, test execution, and zero-downtime deployment on every push to the main branch, cutting deployment cycle time by an estimated 90%.
- Implemented pipeline security best practices including encrypted GitHub Secrets management and scoped IAM permissions for cloud resource access.

Enterprise Network Infrastructure Simulation | Demo | Cisco Packet Tracer, OSPF, VLAN, NAT, DHCP, SSH2024

- Designed and simulated a complete enterprise-grade network topology for a multi-department organization within a virtualized Cisco Packet Tracer environment.
- Configured **VLANs and Inter-VLAN routing** to enforce departmental traffic segmentation, improving security posture and reducing broadcast domain size by 75%.
- Implemented **OSPF dynamic routing protocol** across multiple routers, enabling automatic route convergence and redundant path failover within the simulated WAN.
- Deployed **DHCP** for automated IP allocation, **NAT** for internet address translation, and **SSH** for encrypted remote device management across all network nodes.
- Applied **port security** policies on access-layer switches to prevent unauthorized device connections, simulating real-world threat mitigation scenarios.

AI-Powered Automated Data Analysis Agent | Demo | n8n, Google Sheets API, AI/LLM Integration, Email Automation2025

- Built an end-to-end **AI automation workflow** using n8n that autonomously ingests monthly business data from Google Sheets, performs statistical analysis, and generates visual graphical reports.
- Integrated an **LLM-based AI agent** into the n8n pipeline to produce natural-language data summaries and actionable business insights alongside generated charts.
- Automated the full reporting cycle — data retrieval, analysis, chart generation, and email dispatch — delivering polished monthly reports to stakeholders with zero manual intervention.

Education

Bachelor of Science in Information Technology (B.Sc. IT)	April 2023 – 2026
<i>Pataliputra University — CIMAGE Group of Institutions, Patna, Bihar</i>	GPA: 6.8/68%
12th — Physics, Chemistry, Mathematics (PCM)	July 2020 – March 2022
<i>H.B.J. College Khajauli, Madhubani, Bihar (BSEB)</i>	67%
10th — Secondary Education	Completed July 2020
<i>Awasiya Public School, Madhubani, Bihar (CBSE)</i>	81%

Certifications & Training (certificates available on [portfolio](#))

DevOps+ — CIMAGE Group of Institutions	September 2025
CompTIA A+ & Network+ — CIMAGE Group of Institutions	February 2024
C++ Programming — Spoken Tutorial, IIT Bombay	2023
Python Training — CIMAGE College + Spoken Tutorial, IIT Bombay	2023
Linux System Administration Training — CIMAGE College + Spoken Tutorial, IIT Bombay	2023

Achievements & Additional Information

- Earned **CompTIA A+** and **Network+** internationally recognized certifications, validating expertise in hardware troubleshooting, OS support, and network administration.

- Recognized for **1st Division** academic performance at Graduation, 10th (CBSE) and 12th (BSEB) board examinations.
- Independently designed and deployed multiple full-stack and DevOps projects without formal industry mentorship, reflecting strong self-directed learning and initiative.